Discrete Probability Space

$(\Omega, P): \Omega \neq \emptyset$, $P: P(\Omega) \to [0,1]$

i) $P(\Omega) = 1$ // $\sum_{\omega \in \Omega} P_\omega = 1$

ii) disjoint $A_i \in R$

$P(\bigcup_{i=1}^{n} A_i) = \sum_{k=1}^{n} A_k$

Laplace Probability Space

$P(A) = \frac{|A|}{|\Omega|}$, $P_\omega = \frac{1}{|\Omega|}$

$P(A^c) = 1 - P(A)$

$A \subseteq B \subseteq \Omega$: $P(A) \leq P(B)$

Direct Distribution

$P_\omega = \begin{cases} 1, & \omega = \omega_0 \\ 0, & \omega \neq \omega_0 \end{cases}$

$\sum_{k=0}^{n} a_k \lambda^k = P^\lambda$

Inclusion Exclusion Principle

$P(\bigcup_{k=1}^{n} A_k) = \sum_{k=1}^{n} (-1)^{k-1} \sum_{i_1 < \dots < i_k \leq n} P(\bigcap_{j=1}^{k} A_{i_j})$

$\Omega = \bigtimes_{i=1}^{m} \Omega$ $\quad P(A) = \sum_{(\omega_1, \dots \omega_m) \in A} \prod_{i=1}^{m} P_i(\{\omega_i\})$

Conditional P; A given B

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

Law of total probabilities

$P(A) = \sum_{i \in I} P(A|B_i) P(B_i)$

Bayes Formular

$P(B_i|A) = \frac{P(B_i) P(A|B_i)}{\sum_{j \in I} P(B_j) P(A|B_j)}$

$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$

$(a+b)^m = \sum_{i=0}^{m} \binom{m}{i} a^i b^{m-i}$

Elementary Calibrations

$P(A) = \frac{|A|}{|\Omega|}$

i) Ordered, replacement: $|\Omega| = \mu^n$

ii) Ordered, no replacement: $|\Omega| = \frac{\mu_0!}{(\mu_0 - n)!}$

iii) Unordered, replacement: $|\Omega| = \binom{\mu + n - 1}{n}$

iv) Unordered, no replaced: $|\Omega| = \binom{\mu_0}{n}$

$\binom{\mu}{n} = \frac{\mu_0!}{n_0!(\mu - n)!}$

Independance

$P(A \cap B) = P(A) P(B)$; $P(\bigcap_{i \in I} A_i) = \prod_{i=1}^{n} P(A_i)$

σ-Algebra: $\mathcal{F}$ on Ω, if

(i) $\Omega \in \mathcal{F}$

(ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

(iii) $A_1, \dots, A_n \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

$S(A)$ smallest σ-algebra, $A \subseteq S(A)$

$\Omega = \mathbb{R} \Rightarrow \mathcal{B}(\mathbb{R}) = S(A)$

Probability space $(\Omega, \mathcal{F}, P)$

i) $P(\Omega) = 1$

ii) $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$, A_i disjoint

$X: \Omega \to \mathbb{R}$ random Variable

$P_X(A) = P(\{\omega \in \Omega : X(\omega) \in A\})$

Distribution Function

$P(X \leq y) = F_X(y)$

$P(X = y) = P(X \leq y) - P(X < y) = F_X(y) - F_X(y^-)$

$P(a \leq X \leq b) = F_X(b) - F_X(a^-)$

Density

i) $f(x) \geq 0$

ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\Rightarrow F_X(y) = \int_{-\infty}^{y} f(x)\,dx$

Independance of RV.

$P(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^{n} P(X_i \in A_i)$

Binomial (n,p): $[0, \dots, n]$

$P(X=k) = p_k = \binom{n}{k} p^k (1-p)^{n-k}$

$E[X] = np$ $\quad Var[X] = np(1-p)$

Hypergeometric 66 blue, r red balls, we draw $n \leq b+r$ no replacement

X = # blue balls

$P(X=k) = \frac{\binom{b}{k}\binom{r}{n-k}}{\binom{b+r}{n}}$

$P(A|B) + P(A^c|B) = 1$

Expectation

$E[X] = \sum_{\omega \in \Omega} X(\omega) P(\{\omega\}) = \sum_{y \in X(\omega)} y \cdot P(X=y)$

$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$; $E[R(X)] = \sum_{x \in X(\Omega)} R(x) P(X=x)$

or $= \int_{-\infty}^{\infty} R(x) f(x)\,dx$

$E[X+Y] = E[X] + E[Y]$

$E[aX] = a E[X]$

X, Y independent: $E[XY] = E[X] \cdot E[Y]$

Standarddeviation $\sigma = \sqrt{Var[X]}$

Variance

$Var[X] = E[X^2] - E[X]^2$

$Var[aX] = a^2 Var[X]$

$Var[X+C] = Var[X]$

$Var[X] = Cov(X,X)$

$Var[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} Var[X_i] + 2 \sum_{1 \leq i < j \leq n} cov(X_i, X_j)$

$= 0$ if independ

Covariance

$cov(X,Y) = E[XY] - E[X]E[Y]$

$cov(a_1 X_1 + a_2 X_2, b_1 Y_1 + b_2 Y_2) =$

$a_1 b_1 cov(X_1, Y_1) + a_2 b_1 cov(X_2, Y_1) + a_1 b_2 cov(X_1, Y_2) + a_2 b_2 cov(X_2, Y_2)$

$Y = (\max_{n=1,\dots,n} U_{1,\dots,n}) U_{1,\dots,n}$ Uniform $[0,1]$

$P(Y \leq x) = P(\max_{n=1,\dots,n} U_{1,\dots,n} \leq x) = P(U_1 \leq x)^n$

Cond Expectation

$E[X|A] = \frac{E[X \cdot 1_A]}{P(A)}$ or $E[X|A] = \sum_{y \in X(\Omega)} y P(y|A)$

$E[X] = \sum_{i \in I} P(B_i) E[X|B_i]$

Bernoulli: $X \sim Ber(p)$

$P(X=1) = p$; $P(X=0) = 1-p$

$1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$

$E[X] = p(A)$

Correlation Coefficient

$\rho_{X,Y} = \frac{cov(X,Y)}{\sqrt{Var[X]}\sqrt{Var[Y]}}$

Uniform $(\{a, \dots, b\})$

$F_X(y) = \begin{cases} 0, & y < a \\ \frac{y-a}{b-a}, & y \in [a,b] \\ 1, & y > b \end{cases}$

$E[X] = \frac{a+b}{2}$, $Var[X] = \frac{(b-a)^2}{12}$

Uniform $([a,b])$

$F_X(y) = \begin{cases} 0, & y < a \\ \frac{y-a}{b-a}, & y \in [a,b] \\ 1, & y > b \end{cases}$

$E[X] = \frac{a+b}{2}$, $Var[X] = \frac{(b-a)^2}{12}$

Pareto (α)

$F_X(y) = \begin{cases} 0, & y < 1 \\ 1 - y^{-\alpha}, & y \geq 1 \end{cases}$

Sum of integer valued r.v.

$P(X+Y=n) = \sum_{k=-\infty}^{\infty} P(X=n-k) P(Y=k)$

Moment generating fnkt

$M_X(t) = E[e^{tX}]$

$E[X^m] = M_X^{(m)}(0)$

X, Y independent $\Rightarrow M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$

$\hookrightarrow$ same dist $M_X(t) = M_Y(t)$

Exponential (λ)

$F_X(y) = \begin{cases} 0, & y < 0 \\ 1 - e^{-\lambda y}, & y \geq 0 \end{cases}$

$E[X] = 1/\lambda$, $Var[X] = 1/\lambda^2$

Convolution of densities

$(f * g)(t) = \int_{-\infty}^{\infty} f(t-s) g(s)\,ds$

$f_{X+Y} = f_X * f_Y$

Normal (μ, σ^2)

$\phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-x^2/2}\,dx$

$E[X] = 0$

$Var[X] = 1$

$\phi(-t) = 1 - \phi(t)$

$X \sim Normal(\mu, \sigma)$

$X - \mu \sim Nord(0, \sigma)$

$\frac{X-\mu}{\sigma} \sim Nord(0,1)$

$X + Y \sim Nord(\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2})$

$aX \sim Normal(a\mu, a\sigma)$

$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i \sim Normal(\mu, \sigma/\sqrt{n})$

Weak law of large numbers

$(X_i)_{i \in \mathbb{N}}$ be a sequence of i.i.d. r.v. such that $E[X_1^2] < \infty$ and $S_n = \sum_{i=1}^{n} X_i$ for $n \in \mathbb{N}$

$\frac{S_n}{n} \xrightarrow{P} E[X_1]$, as $n \to \infty$

Markov Inequality: $P(|X| \geq a) \leq \frac{E[R(X)]}{R(a)}$

Chebyshev Inequality: $P(|X - E[X]| \geq a) \leq \frac{Var[X]}{a^2}$

$(X_i)_{i \in \mathbb{N}} \sim Ber(p)$ $\quad \frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{P} p$ as $n \to \infty$

Convergence in Probability

$(X_n)_{n \in \mathbb{N}}$ converges in probability to X if

$\lim_{n \to \infty} P(|X_n - X| \geq \varepsilon) = 0$,

$X_n \to X$, as $n \to \infty$

Central Limit Theorem

$S_n = \sum_{i=1}^{n} X_i$

$\lim_{n \to \infty} P\left(\frac{S_n - n E[X_n]}{\sqrt{n}\sqrt{Var[X_1]}} \leq t\right) \approx \phi(t)$

$Y_n \sim Binom(n,p)$

$P(Y_n \leq a) = P\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \leq \frac{a - np}{\sqrt{np(1-p)}}\right) \approx \phi(**)$

Convergence in Distribution

$\lim_{n \to \infty} F_{X_n}(t) = F_X(t)$

Poisson Process

$\lambda > 0$ and iid $Exp(\lambda) \sim r.V.(w_i)_{i \in \mathbb{N}}$

$X_n = \sum_{i=1}^{n} w_i$

$N_t = \sum_{n=1}^{\infty} 1_{\{X_n \leq t\}}$, $t \geq 0$

$s \leq t$

$N_t - N_s \sim Poisson(\lambda(t-s))$

Brownian Motion

(i) $P(B_0 = 0) = 1$

(ii) $P(t \mapsto B_t$ continuous$) = 1$

(iii) $(B_t)_{t \geq 0}$ has independent increments

(iv) For $s, t \geq 0$, $s \leq t$

$B_t - B_s \sim Normal(0, \sqrt{t-s})$

$B_t \sim Normal(0, \sqrt{t})$

Independent Increments

$X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}}$ are independant

Stochastic Process:

a collection $(X_t)_{t \in T}$ with values in E

$T \ni t \mapsto X_t(\omega)$

mean function
$\mu(t) = E[X_t]$
covariance function
$\rho(s,t) = \mathrm{cov}(X_s, X_t)$

Markov Property $\Rightarrow$ Markov Chain $(X_t)_{t\in\mathbb{N}_0}$
$P(X_t = i \mid X_{t_n} = i_n, \dots, X_{t_1} = i_1) = P(X_t = i \mid X_{t_n} = i_n)$
transition probs
$P(X_{t+1} = j \mid X_t = i) = p_{ij}$
$P(X_{t+1} = j) = \sum_{i\in\mathbb{E}} p_{ij} P(X_t = i)$

$P = \begin{pmatrix} p_{11} & \cdots & p_{1m} \\ p_{m1} & \cdots & p_{mm} \end{pmatrix}$
$p(t) = (P(X_t = 1), \dots, P(X_t = m))$
$p(t+1) = p(t) \cdot P$

Point estimates
$T : X \rightsquigarrow \mathbb{R}$
unbiased if $E_\vartheta[T] = \tau(\vartheta)$
bias $B_\vartheta(\vartheta) = E_\vartheta[T] - \tau(\vartheta)$

Sample mean: $\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i$
sample variance: $s_x^2 = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})^2$
sample median
$m_x = \begin{cases} x_{n+1/2} & n \text{ is odd} \\ \frac{x_{n/2} + x_{n/2+1}}{2} & n \text{ is even} \end{cases}$
$P(X \geq m) \geq \frac{1}{2} \geq P(X \leq m)$

Statistical Model
$(\mathcal{X}, \mathcal{F}, (P_\vartheta)_{\vartheta\in\Theta})$
$\mathcal{X}$ sample space, $\mathcal{F}$ σ-algebra, probability measures $\Rightarrow$ joint distribution if n iid r.v

One sided Gauss Test:
$(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (P_\mu)_{\mu\in\mathbb{R}})$
$x_1, \dots, x_n \sim \mathrm{Normal}(\mu, \sigma_0)$
$H_0 : \mu \leq \mu_0$ & $H_1 : \mu \geq \mu_0$
$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \bar{x} \leq \mu_0 + z_{1-\alpha}\frac{\sigma_0}{\sqrt{n}} \\ 1, & \bar{x} \geq \mu_0 + z_{1-\alpha}\frac{\sigma_0}{\sqrt{n}} \end{cases}$

Two sided Gauss
$H_0 : \mu = \mu_0$
$H_1 : \mu \neq \mu_0$
$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \bar{x} \in (\mu_0 - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu_0 + z_{1-\alpha/2}\sigma_0/\sqrt{n}) \\ 1, & \bar{x} \notin (-\!\!|\!\!-) \end{cases}$

Consistency
$\lim_{n\to\infty} P_\vartheta(|T_n - \tau(\vartheta)| \geq \varepsilon) = 0$
$\Leftrightarrow T_n \xrightarrow{P} T$ as $n \to \infty$
$T_n = \frac{1}{n}\sum_{i=1}^n x_i \xrightarrow{P_\vartheta} E_\vartheta[X_1]$

eg: Confidence Intervall: $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (P_\mu)_{\mu\in\mathbb{R}})$
$x_1, \dots, x_n \sim \mathrm{Normal}(\mu, \sigma)$
$P_\mu(\{x \in \mathcal{X} : \mu \in (\bar{x} - c, \infty)\}) \geq 1 - \alpha$
$= P_\mu(\bar{x} - c \leq \mu) = P_\mu\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq \frac{c}{\sigma/\sqrt{n}}\right)$
$= \Phi\left(\sqrt{n}\frac{c}{\sigma}\right) \geq 1 - \alpha$
$\Rightarrow \sqrt{n}\frac{c}{\sigma} \overset{!}{=} z_{1-\alpha}$
$\Rightarrow \Theta = [\bar{x} - \frac{z_{1-\alpha}}{\sqrt{n}}\sigma, \infty)$

eg: Likelihood
$\mathcal{L}_x(p) = P_p(\{x\}) = P_p(x_1 = x_1, \dots, x_n = x_n)$
$= P_p(x_1 = x_1) \cdot \dots \cdot P_p(x_n = x_n)$

eg: consistency
$\frac{1}{n}\sum_{i=1}^n x_i \xrightarrow{P} E[x_1] = m$
$\hat{\sigma}_n^2 = \frac{1}{n-1}\sum_{i=1}^n x_i = \frac{n}{n-1}\cdot\frac{1}{n}\sum_{i=1}^n x_i \xrightarrow{P} m$
$n \to \infty \to 1$

Maximum Likelihood Estimator
$P_{T(x)}(\{x\}) = \max_{\vartheta\in\Theta} P_\vartheta(\{x\})$
$\hookrightarrow$ max bei ϑ
$\mathcal{L}_x : \Theta \ni \vartheta \mapsto P_\vartheta(\{x\}) = P_\vartheta(x_1 = x_1, \dots, x_n = x_n)$

Confidence Interval
$C : \mathcal{X} \to \mathcal{I}$
$P_\vartheta(\{x \in \mathcal{X} : \tau(\vartheta) \in C(x)\}) \geq 1 - \alpha$
error level α (confidence lvl $1-\alpha$)

Quantile
p-quantile $P(x \overset{\leq}{\geq} q_p) \geq p$ &
$P(x \geq q_p) \geq 1 - p$

Test: $H_0 : \vartheta \in \Theta_0$
$H_1 : \vartheta \in \Theta_1$
$\varphi : \mathcal{X} \to \{0, 1\}$
if $\varphi(x) = 1$, reject H_0
if $\varphi(x) = 0$, do not reject H_0
level α if
$P_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}) \leq \alpha$
Power of $\varphi(x)$

One Sided Binomial Test
$(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (P_p)_{p\in[0,1]})$
P_p is joint Dist of iid
$x_1, \dots x_n \sim \mathrm{Ber}(p)$
$H_0 : p \leq p_0$ / $H_1 : p > p_0$
$m \in \mathbb{N}$
$\varphi : \{0,1\}^n \ni x \mapsto \begin{cases} 0, & \sum_{i=1}^n x_i \leq m_0 \\ 1, & \sum_{i=1}^n x_i > m_0 \end{cases}$
$p \leq p_0$